Received: 29 June 2017 | Accepted: 14 August 2017

DOI: 10.1111/2041-210X.12893

RESEARCH ARTICLE

Methods in Ecology and Evolution BRITISH ECOLOGICAL SOCIETY

An open-source sensor-logger for recording vertical movement in free-living organisms

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Handling Editor: Nicolas Lecomte

Abstract

1. Measuring individual-based movements in free-ranging animals is relevant to our understanding of many basic ecological principles (migration, competition, optimal foraging) but has been difficult to realize in a small size.
2. We have developed a freely available, open-source, non-invasive datalogger that can measure an animal's altitude or elevation (A) at sampling intervals greater than once per second, storing up to 10,000 total samples, depending on programme size.
3. The realized design weighs 370 mg, opening a wealth of opportunities to study animals as small as 10 g without exceeding the 4% rule-of-thumb for the relative mass of telemetry devices.
4. In addition, power management is provided via a modern ultra-low-power microcontroller, making it possible to record altitude data over extended periods (>100 days) by extending battery lifetimes.
5. First results from testing and field applications of this design underline its suitability for reliable and extended monitoring of how animals interact with the environment. Our empirical results show this design can provide high-resolution data with altitude measurements accurate to the nearest several metres at 1-s intervals, provided a local known elevation pressure reference, elucidating fine-scale differences between species/individuals in their use of the air column over 10-day intervals.

KEYWORDS

altitude, ambient temperature, animal movement, foraging behaviour, migration, open-source, telemetry

1 | INTRODUCTION

The lives of vertebrates are intimately connected to their environment; they use the space above and around them to forage, mate, migrate, avoid predators and disperse (Oresman et al., 2013). Movement through the environment is the central hub connecting these various functions throughout an animal's life span, and movement abilities reflect the ecological and evolutionary histories of different lineages (Hansson & Åkesson, 2014). For example, the advent of flight launched vertebrates into a new habitat—the airspace—leading to a rapid adaptive radiation of life history and behavioural traits as they filled a niche previously occupied only by insects (Dudley & Yanoviak, 2011). Exploiting the vertical dimension of habitat to an extreme, some volant animals spend almost their entire lives airborne, returning to land only in brief bouts during a short breeding season to rear young (Hedenström et al., 2016; Liechti, Witvliet, Weber & Bächler, 2013). The vertical aspect of movement is not limited to the airspace, and habitat selection in the terrestrial realm ranging from a single tree in *Anolis* lizards (Kamath & Losos, 2017; Losos et al., 2003; Pasch, Bolker & Phelps, 2013) to altitudinal bands along a mountain range in Neotropical singing mice (*Scotinomys*, Pasch et al., 2013) underlies adaptive radiations and species boundaries. Clearly, movement is fundamental to many ecological questions and principles, and data are often only informative when obtained from the organismal perspective.

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To understand how vertebrates interact with their environment requires an integrated approach of documenting organismal behaviour and characteristics of the environment. Recent advances in environmental sensing technology have helped researchers overcome some of these challenges; for example, global positioning system (GPS) and very-high frequency (VHF) radiotelemetry technology have provided a more refined view of species- and individual-level variation in movement behaviour (Kays, Crofoot, Jetz & Wikelski, 2015; López-López, 2016; Wilmers et al., 2015). Fundamental aspects of animal biology and migration are still being uncovered from telemetry-based studies of individual organisms, from great frigatebirds (*Fregata minor*) sleeping during flight using a single brain hemisphere (Rattenborg et al., 2016), to the marathon 10-month continuous airborne lifestyle of the swifts (Hedenström et al., 2016; Liechti et al., 2013). The recency of these discoveries regarding different dimensions of habitat use and basic science highlights that we are on the verge of a new era for understanding animal movement from an organismal perspective.

Despite the rapid development of technology that can be tailored to obtain movement data from free-living animals, its use by biologists continues to lag. It is likely that the high cost of commercial tracking technology is a major factor retarding its adoption and widespread implementation in research (MacCurdy et al., 2009). High unit costs are intrinsic to the custom-designed electronics frequently used by biologists, but lower cost—though no less effective—solutions may now be within reach of biologists with moderate to low budgets (Allan, Arnould, Martin & Ritchie, 2013; Forin-Wiart, Hubert, Sirguey & Poulle, 2015; Quaglietta, Martins, de Jongh, Mira & Boitani, 2012). The explosion of consumer demand for cellular and Internet-of-Things (IoT) technologies has promoted the development of low-cost microelectromechanical (MEM) sensors and small, low-power microcontrollers. Although they were not developed for use in biological research, the small size of these commercial off-the-shelf environmental sensors and processors is ideal for the study of organismal movement, where minimizing mass is of paramount importance. Previous studies have shown that radiotelemetry or datalogging devices should not exceed 3%–5% of a flying (Aldridge & Brigham, 1988) or 5%–10% of a terrestrial vertebrate's mass (Sikes, 2016), as exceeding these limits can influence the behaviour and even survival of tagged animals (Barron, Brawn & Weatherhead, 2010). Integrating low-cost environmental sensors into the most compact package possible will allow researchers to safely record novel data from a broader range of species, providing more comprehensive insights into both the environmental conditions vertebrates experience and their behavioural responses to those conditions.

Here, we present an open-source design for a simple, lightweight (370 mg) barometric pressure datalogger that can record time and atmospheric pressure over several weeks to multiple months for the study of flying animals as small as 12.5 g (*c*. 4% body mass) and terrestrial animals as small as 4 g (*c*. 10% body mass). Barometric pressure can be converted into measures of altitude provided local known elevation pressure readings, recording relevant data on how animals use the vertical dimension of habitat. The datalogger was designed to use a minimum of components to save on mass and complexity, resulting in a finished product using only seven total components (Figure 1, Table 1). We provide all the necessary design files and code for a person with a minimal background in electrical engineering to produce the field-tested reference design in this paper (https://github.com/trollock/OS-Altitude-Datalogger). It is our hope that interested researchers will use and continue to refine and adapt the technology we present here under an open-source framework.

2 | MATERIALS AND METHODS

2.1 | Datalogger design overview

Each datalogger consists of a BME280 pressure, temperature and humidity sensor (Bosch Sensortec, Reutlingen, Germany), a MSP430FR5949 microcontroller (MCU) with 64-Kb non-volatile ferromagnetic random

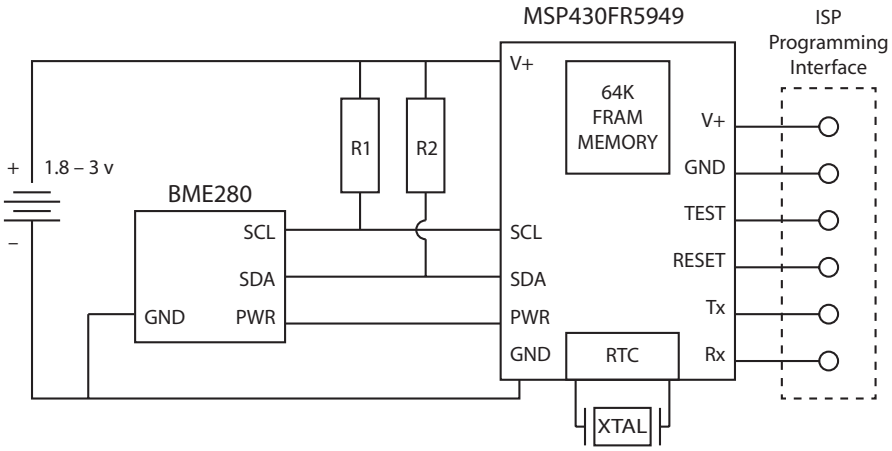


FIGURE 1 The datalogger electrical diagram—the final datalogger design includes only seven total components, weighing a minimum of 370 mg (with 3.4-mAh battery, crystal load capacitors not shown for clarity). Input power to the sensor (BME280) is controlled by the microcontroller (MCU—MSP430FR5949), allowing for maximum power savings during low-power modes. The data communication between sensor and MCU uses an I2C bus, with two pull-up resistors. Keeping track of time between sampling intervals is controlled by an internal real-time clock (RTC) and an external crystal (XTAL)

TABLE 1 Datalogger components and specifications

Component	Manufacturer	Part number/value	Physical size	Package	Mass (mg)
Microcontroller	Texas instruments	MSP430FR5949	6 × 6 mm	40-pin VQFN	82
Pressure sensor	Bosch GMBH	BME280	2 × 2.5 mm	8-pin LGA	23
External crystal	Ambiq	ABS-06	2 × 1 mm	2-SMD	18
Capacitor (X2)	Generic	12.5 pF	0.2 × 0.2 mm	0201	2
Resistor (X2)	Generic	4.7–10 kΩ	0.2 × 0.1 mm	0201	2
Battery	Seiko	MS614SE	6.8 mm dia.	FL28N	165
Circuit board	Multiple vendors	N/A	8 × 9 mm	0.1 mm THK	47
				Device mass	339
UV and water-resistant epoxy (aerosol)				Total mass	371
UV and water-resistant marine epoxy (gel)				Total mass	483

access memory (FRAM, Texas Instruments), and an external 32.748-kHz crystal oscillator to minimize clock drift (Figure 1). For our tests, we chose an intermediate-sized rechargeable manganese lithium battery (3.1 V, 3.4 mAh, Seiko Instruments Inc, Chiba, Japan). The entire device was weatherproofed with one of two methods (to test for mass vs. durability, Table 1), (1) spray coated with clear UV- and water-resistant epoxy (Krylon, Cleveland, OH, USA) or (2) dipped in thinned marine epoxy (Loctite Henkel Corporation, Düsseldorf, Germany) with the aperture opening to the BME280 sensor covered with 2 × 2 mm adhesive-backed Gore-tex to allow measurement of air pressure while protecting the sensor from moisture (Figure 2). The device has six contacts for programming and data retrieval via ISP (inline serial programming). We removed excess weatherproofing (e.g. epoxy) from these contact points to ensure a good electrical connection during data retrieval.

2.2 | Datalogger Life Expectancy

We measured the power consumption from five dataloggers in each of the power modes (active mode, low-power mode) to calculate the power consumption of different duty cycles. Detailed information on other sampling rates, projected life spans and memory allotment is provided in Data S1.

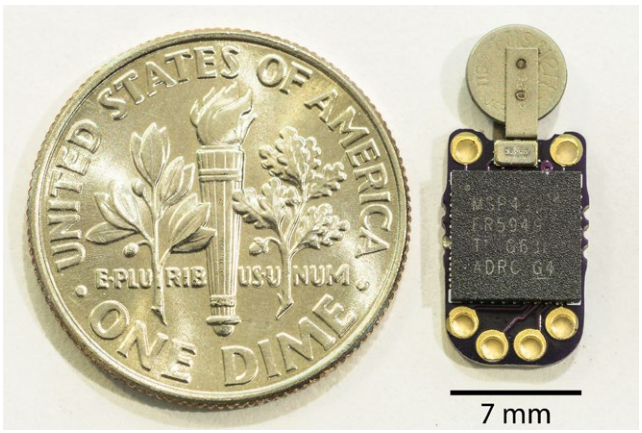


FIGURE 2 Photograph of the device weighing *c*. 370 mg next to a US dime that is approximately 17.9 mm across. The list of components and specifications are provided in Table 1. Scale bar is equal to 7 mm

2.3 | Clock drift

We determined the drift of the real-time clock using five dataloggers by recording on-board clock time at the beginning and end of an interval of at least 24 hr at 25°C and comparing the on-board time to measured time.

2.4 | Altitudinal accuracy (known reference elevation pressure)

Altitudinal accuracy of the dataloggers was determined by taking five individual dataloggers to several locations of known altitude simultaneously, holding them at a location for 10 min, and comparing the recorded pressure values from each of the locations. Ground truth atmospheric pressure values were from the Ithaca Airport AWOS (Automated Weather Observation System) station, which is no more than 5 km from any of the altitude test sites.

2.5 | Altitudinal accuracy (estimated sea-level pressure)

We calculated altitudinal accuracy for situations where the estimated sea-level pressure was within 100, 500, 1,000 and 5,000 Pa of real sea-level pressure using the altitude formula listed below.

2.6 | Field testing

We attached dataloggers to purple martins (*Progne subis*, PUMA), tree swallows (*Tachycineta bicolor*, TRES) and barn swallows (*Hirundo rustica*, BARS) at three different sites with nest boxes or breeding colonies in Tompkins and Schuyler Counties, NY, between 22 May 2016 and 20 July 2016. The sites were at private residences located on Seneca Lake (PUMA, 42.388116 N, −76.877739 W, elevation 130 m) and Danby (BARS, 42.287893 N, −76.454121 W, elevation 395 m), and a long-term study site (TRES, 42.504130 N,

−76.463413 W, elevation 335 m). These three sites are within 25 km of each other.

2.7 | Estimating foraging altitudes

Birds were captured at known nest sites, and the dataloggers were mounted using a Rappole-style harness (Rappole & Tipton, 1991) made from 0.7-mm Stretch Magic (Pepperell Braiding Company, MA, USA) elastic cord. The dataloggers were programmed to record air pressure and temperature every 60 s over a period of 3–15 days (depending on our ability to retrieve the dataloggers at the nest site). When a tag was retrieved from a tagged bird, the data were downloaded over the six-pin ISP interface that connected a serial terminal to the power and communication terminals of the datalogger. Because atmospheric pressure varies temporally, datalogger information was drift-corrected against pressure-level readings from the nearest airport AWOS station at the Ithaca Airport, Tompkins County, NY (42.504130 N, −76.463413 W, elevation 335 m). Using the local AWOS weather data as ground truth, we converted the logger data to altitude using the following rearranged version of the barometric formula to obtain altitude.

$$A = \frac{\left(\left(\frac{P_0}{P}\right)^{\frac{1}{5.257}} - 1\right)(T + 273.15)}{0.0065}$$

where A = datalogger altitude (in metres), P_0 = sea-level pressure (in pascals), P = datalogger pressure (in pascals) and T = temperature at sea level (in Kelvin).

3 | RESULTS

Our datalogger design (v3.0.1) using aerosol UV-resistant epoxy weatherproofing weighs 370 mg (Table 1) and when combined with a Rappole harness fitted for a tree swallow, weighed *c.* 550 mg. Using a 3.4-mAh battery and collecting samples at an interval of once per 300 s (1/300 Hz, average current 3.40 μA), the datalogger life span is an estimated 40 days when using measured current draw and memory allotment.

The clocks on individual tags showed a mean drift of 23 ± 5 ppm, or a mean drift of 1.99 ± 0.43 s/day. All tested clocks on dataloggers were slower than measured time.

After being converted to altitude in metres, the mean difference between the true altitude (from USGS National Elevation Dataset, NED) and the datalogger was 1.4 m ± 1.1 m. Using estimates of sea-level pressure that were 100, 500, 1,000 and 5,000 Pa greater than the true sea-level pressure values resulted in calculations of altitude that were 8.56 ± 0.05, 42.72 ± 0.27, 85.29 ± 0.54 and 419.88 ± 2.70 m different from actual altitude.

Downloaded data from one of the field-tested designs in Figure 3 illustrate a variety of potential field applications depending on the experimental design or question. By recording altitude and thus flight behaviour, we can determine relative activity for each individual throughout their daily cycle (Figure 3a), the aggregate of this behaviour over multiple days (Figure 3b) and the altitudinal distribution over time (Figure 3c).

4 | DISCUSSION

Over the last decade, we have seen the development of movement ecology as a discipline seeking to explore how animals interact with all dimensions of their habitat (Nathan et al., 2008). Despite a rich history, much of what we know about movement behaviour in all terrestrial or aerial vertebrates comes from limited, snapshot, ground-based observations. As the size and weight of telemetric tools shrink, our understanding of how vertebrates use the vertical dimension will transition to measures taken directly on the moving animal, rather than estimated by a ground-based observer. Here, we present a simple, low-cost, open-source design that can provide accurate and precise altitudinal data on the vertical aspect of organismal movement.

4.1 | Selection of temporal resolution

The ability for researchers to choose among different levels of temporal resolution and battery life makes our datalogger a robust tool for answering both proximate and ultimate questions about the ecology of aerial organisms. For instance, high sampling rates (greater than once per second) over shorter time-scales can provide almost continuous, high-resolution data for within-season behaviours (e.g. displays/movement/locomotion (flight behaviour for instance, Figure 3), and if applicable, foraging). In contrast, less frequent sampling (once per minute or less) over longer time-scales can improve our understanding

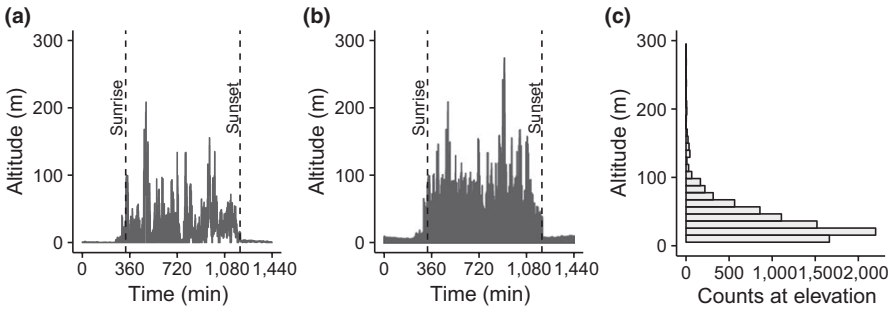


FIGURE 3 Sample data for foraging altitude of an individual barn swallow. (a) Foraging altitude at minute intervals over a single day. (b) Composite of foraging altitudes over a 5-day interval (18–22 June). (c) Distribution of time spent at different altitudes from data sampled over 5 days. All data are sampled at an interval of once per minute

of longer-term movement, habitat affiliations and migration strategies in study taxa.

Two of the most important aspects to consider when using these data are (1) how the sampling schedule influences interpretation and analysis of data collected and (2) the ability to gather reference elevation pressure information. The data on foraging of a single barn swallow summarized in Figure 3 display the first challenge. In our sample data, we recorded instantaneous pressure readings on a regular schedule of once per minute. As individual nest visits are often very brief, these events are less likely to be captured at this sampling interval and are likely underrepresented in the data. Threshold-dependent schemes (such as having the datalogger only record samples when the datalogger is above a certain pressure level or the deviation from a running average) are possible solutions depending on the specific application and question. The second challenge is evidenced by the altitude calculations where the reference elevation pressure is estimated. On a typical day without severe weather, the atmospheric pressure can change over 500 Pa, and using a single average pressure value for that day could result in a miscalculation of altitude by as much as 42 m. This underscores the importance of designing studies where accurate local pressure information can be measured directly or interpolated from weather services.

4.2 | Research areas that could be advanced with altitudinal logger data

4.2.1 | Studies of animal competition

Individual-based measurements of altitude can help ecologists measure patterns of both intra- and interspecific competition. Within a species, competitive foraging strategies related to aerial behaviour can be tied to individual factors such as body condition, personality, dominance status, experience, sex, age, and breeding stage, as well as to environmental conditions such as levels of food availability and ambient temperatures and atmospheric conditions. At the level of communities, individual-based aerial sensor data can be used to identify aerial/altitudinal stratification of foraging zones in areas of sympatry between ecologically similar species. Such data can also reveal the temporal dynamics of stratification as community composition changes by providing information on individual-based responses as in Helms, Godfrey, Ames and Bridge (2016), using methods such as those presented here to overcome the limitations of coarser-scale tools such as radar.

4.2.2 | Flight dynamics during migration

While geolocators have revealed the position of migratory birds in the x–y plane at broad spatial scales (Beason, Gunn, Potter, Sparks & Fox, 2012), we largely lack information on their movements with respect to the "z" or azimuthal dimension in small species, as altimeters/barometric sensors to date have only been available on relatively large tags (Burger & Shaffer, 2008; Weimerskirch, Chastel, Barbraud & Tostain, 2003). Provided a method to estimate a local reference elevation pressure, sampling during migration and the non-breeding season could shed light on whether daily flight/foraging patterns change over the annual cycle for many species and can also reveal, as with accelerometers, how much time is spent on the wing vs. roosting (e.g. figure 3 from Liechti et al., 2013). Fine-scale measurements of individual flight movements can also reveal species-level differences in the dynamics of flight behaviour. For instance, barometric pressure data can be used to investigate how individuals/species use thermals and interact with storm fronts and tail/headwinds during migration, providing an important complement to coarse population-level estimates of flight behaviour that have been made to date (Horton et al., 2016; La Sorte et al., 2014).

4.2.3 | Altitudinal migration

Our barometric datalogger can also be used to investigate the migration behaviour of both aerial and terrestrial altitudinal migrants. Our understanding of altitudinal migration is based largely on species presence/absence data from surveys or changes in capture rates (Boyle, Norris & Guglielmo, 2010). Comparatively little is understood about the individual factors that determine the timing and spatial patterns of altitudinal migration. High-precision, accurate and time-stamped altitudinal data can help determine the causal forces that underlie movement choices, and this clearly may have far-reaching implications for conservation efforts (see discussion below).

4.2.4 | Behavioural repertoires and individual variation

Telemetered accelerometer data have been used to understand time budgets of different classes of behaviour (e.g. foraging, travelling, immobile), but such data often require complicated validation techniques to reliably extract information about specific behavioural states (Hammond, Springthorpe, Walsh & Berg-Kirkpatrick, 2016). By combining elevation data with accelerometry data, this would provide an additional mode of information for validating behavioural state analyses. Furthermore, altitudinal information is an absolute positional measure (with reference and calculated error) that does not depend on movement, whereas data provided by accelerometers are necessarily relative and require animals to be mobile and changing velocity (e.g. the animal may be moving vertically, or horizontally, but its absolute location in vertical space is unknown). The additional information provided by a barometric logger could help in separating behavioural state data if certain behaviours (e.g. foraging on pine cones) are performed only at certain altitudes.

4.2.5 | Conservation efforts

The technology presented here also has the potential to assist in developing conservation-oriented predictive models for the daily and seasonal movements of aerial and terrestrial vertebrates with respect to meteorological, ecological and anthropogenic parameters (Kays et al., 2015). By establishing a functional window into the behavioural

ecology of vertebrate taxa, we can gain a much stronger understanding of the dynamics of the ecosystem and their movements. For aerial vertebrates, the threats to this ecosystem are widespread; for instance, populations of many species of aerial insectivores have declined over the past 25–50 years in North America, and the potential causes for this are diverse and poorly understood (Oresman et al., 2013; Sauer et al., 2014). Conservation benefits of predictive models range from mitigating wind turbine strikes by altering operational schedules during periods of high activity to informing "lights out" events to prevent collisions with high-rise structures (Shipley, Kelly & Frick, 2017). For both aerial and terrestrial organisms, we may potentially begin to forecast how the ecology and their movements will likely change in response to a variety of global perturbations such as warmer temperatures, more pronounced seasonality, and changes in prey abundance and diversity. This information may help policymakers decide where to establish protected areas such as movement corridors and biodiversity hotspots. Protecting and ensuring the continued function of these dynamic ecosystems and their mobile inhabitants in the Anthropocene will require a comprehensive understanding of their ecology, something we believe the use of technology we present here can help to achieve.

AUTHORS' CONTRIBUTIONS

J.R.S. and J.V.K. conceived the study; J.R.S., R.A.D. and D.W.W. designed and performed the experiments; J.R.S. and R.A.D analysed the data; J.R.S., J.V.K., R.A.D. and D.W.W. wrote the manuscript.

ACKNOWLEDGEMENTS

This research was supported by funds from the National Science Foundation (Grant Numbers DBI-1556138 and DEB-1242573 to DWW), Cornell University's Atkinson Center for Sustainable Biodiversity, and the Cornell Laboratory of Ornithology. We would like to thank Cornelia Twining and three anonymous reviewers that greatly improved previous versions of this manuscript.

DATA ACCESSIBILITY

All data, design files, software and related information are openly available from the GitHub data archive located at https://doi.org/10.5281/zenodo.858673.

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SUPPORTING INFORMATION

Additional Supporting Information may be found online in the supporting information tab for this article.

How to cite this article: Shipley JR, Kapoor J, Dreelin RA, Winkler DW. An open-source sensor-logger for recording vertical movement in free-living organisms. *Methods Ecol Evol*. 2018;9:465–471. https://doi.org/10.1111/2041-210X.12893